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Chapter 1

Idempotent Elements in Finite Semigroups

Lemma 1 (Existence of Idempotent Powers). *In a finite semigroup, for any element x , there exists a positive integer m such that x^m is an idempotent. This idempotent power is unique.*

Proof. Since the semigroup S is finite, for any $x \in S$, the set of its powers $\{x^1, x^2, x^3, \dots\}$ must also be finite. By the pigeonhole principle, there must exist distinct positive integers m, n with $m < n$ such that $x^m = x^n$. This gives us $x^m = x^m \cdot x^{n-m}$, showing that powers of x eventually enter a cycle of length $\ell = n - m$ after an initial “tail” of length m .

To construct an idempotent, we choose the exponent $k = 2m\ell$. This exponent is chosen to satisfy two key properties: (1) it is beyond the pre-period m , ensuring we are in the cyclic part, and (2) it is a multiple of the cycle length ℓ . Since $k > m$ and adding multiples of ℓ to an exponent beyond m does not change the result, we have:

$$(x^k)^2 = x^{2k} = x^{k+k} = x^{k+2m\ell} = x^k$$

where the last equality follows from the loop property applied with $2m$ additional cycles of length ℓ . Thus x^k is idempotent.

For uniqueness, suppose x^a and x^b are both idempotent. Since idempotent elements satisfy $e^n = e$ for all positive n , we have:

$$x^a = (x^a)^b = x^{ab} = (x^b)^a = x^b$$

Here we use that x^a raised to the power b equals x^{ab} (by associativity and the idempotent property), and similarly for x^b raised to the power a . Since both expressions equal x^{ab} , the two idempotent powers must be equal. $\square$

Lemma 2 (Sandwich Property in Finite Monoids). *In a finite monoid M , if an element a satisfies the property $a = xay$ for some $x, y \in M$, then there exist non-zero powers of x and y , say n_1 and n_2 , such that $x^{n_1}a = a$ and $ay^{n_2} = a$.*

Proof. From $a = xay$, we can repeatedly substitute a into itself to get $a = x^k ay^k$ for any $k \geq 1$. Since M is a finite monoid, there exists a non-zero power n_1 such that x^{n_1} is an idempotent (by 1). Then we have $a = x^{n_1} ay^{n_1}$. Multiplying by x^{n_1} on the left gives $x^{n_1} a = x^{2n_1} ay^{n_1}$. Since x^{n_1} is idempotent, $x^{2n_1} = x^{n_1}$, so $x^{n_1} a = x^{n_1} ay^{n_1} = a$. A symmetric argument shows that $ay^{n_2} = a$ for some idempotent power y^{n_2} . $\square$

Chapter 2

Green's Relations

Definition 3 (Green's Preorder and Equivalence Relations). In a semigroup S , Green's relations are a set of five equivalence relations that characterize the elements of S in terms of the principal ideals they generate. These relations are fundamental to the study of semigroups.

First, we define four preorder relations: $\leq_{\mathcal{R}}$, $\leq_{\mathcal{L}}$, $\leq_{\mathcal{J}}$, and $\leq_{\mathcal{H}}$. Let S^1 be the monoid obtained by adjoining an identity element to S if it does not already have one. For any two elements $x, y \in S$:

- $x \leq_{\mathcal{R}} y$ iff the principal right ideal generated by x is a subset of the principal right ideal generated by y ($xS^1 \subseteq yS^1$). In Lean, we use the equivalent definition that there must exist some $z \in S^1$ such that $x = yz$.
- $x \leq_{\mathcal{L}} y$ iff the principal left ideal generated by x is a subset of the principal left ideal generated by y ($S^1x \subseteq S^1y$). In Lean, we use the equivalent definition that there must exist some $z \in S^1$ such that $x = zy$.
- $x \leq_{\mathcal{J}} y$ iff the principal two-sided ideal generated by x is a subset of the principal two-sided ideal generated by y ($S^1xS^1 \subseteq S^1yS^1$). In Lean, we use the equivalent definition that there must exist some $u, v \in S^1$ such that $x = uyv$.
- $x \leq_{\mathcal{H}} y \iff x \leq_{\mathcal{R}} y \text{ and } x \leq_{\mathcal{L}} y$.

These four relations are preorders (they are reflexive and transitive).

From these preorders, we define the equivalence relations $\mathcal{R}$, $\mathcal{L}$, $\mathcal{J}$, and $\mathcal{H}$ as the symmetric closures of their corresponding preorders. For example, $x\mathcal{R}y \iff x \leq_{\mathcal{R}} y \text{ and } y \leq_{\mathcal{R}} x$. The equivalence classes are denoted by $[x]_{\mathcal{R}}$, $[x]_{\mathcal{L}}$, etc.

Finally, the $\mathcal{D}$ relation is defined by composing $\mathcal{R}$ and $\mathcal{L}$: $x\mathcal{D}y$ if there exists an element $z \in S$ such that $x\mathcal{R}z$ and $z\mathcal{L}y$. It can be shown that $\mathcal{D}$ is an equivalence relation and that it can also be defined by composing $\mathcal{L}$ and $\mathcal{R}$.

Lemma 4 (Multiplication Compatibility of Green's Relations). *The $\mathcal{R}$ and $\mathcal{L}$ relations exhibit compatibility with semigroup multiplication on one side. Specifically, the $\mathcal{R}$ -preorder is compatible with left multiplication, and the $\mathcal{L}$ -preorder is compatible with right multiplication. If $x \leq_{\mathcal{R}} y$, then for any $z \in S$, we have $zx \leq_{\mathcal{R}} zy$. This property extends to the equivalence relation $\mathcal{R}$. If $x\mathcal{R}y$, then $zx\mathcal{R}zy$. A similar argument holds for the $\mathcal{L}$ -preorder and $\mathcal{L}$ -equivalence, which are compatible with right multiplication. If $x \leq_{\mathcal{L}} y$, then $xz \leq_{\mathcal{L}} yz$ for any $z \in S$.*

Proof. Let $x, y, z \in S$. To prove left compatibility for $\leq_{\mathcal{R}}$, assume $x \leq_{\mathcal{R}} y$. By definition, there exists $a \in S^1$ such that $x = ya$. Multiplying by z on the left gives $zx = z(ya) = (zy)a$. This implies $zx \leq_{\mathcal{R}} zy$. For the equivalence $x\mathcal{R}y$, we have both $x \leq_{\mathcal{R}} y$ and $y \leq_{\mathcal{R}} x$. Applying the result for preorders, we get $zx \leq_{\mathcal{R}} zy$ and $zy \leq_{\mathcal{R}} zx$, which means $zx\mathcal{R}zy$. The proof for $\leq_{\mathcal{L}}$ and $\mathcal{L}$ with right multiplication is analogous. $\square$

Lemma 5 (Commutation of R and L Relations). *The composition of the relations $\mathcal{R}$ and $\mathcal{L}$ is commutative. That is, for any $x, y \in S$, there exists a z such that $x\mathcal{R}z$ and $z\mathcal{L}y \iff$ there exists a w such that $x\mathcal{L}w$ and $w\mathcal{R}y$. This can be written as $\mathcal{R} \circ \mathcal{L} = \mathcal{L} \circ \mathcal{R}$. This property is crucial for proving that the relation $\mathcal{D} = \mathcal{R} \circ \mathcal{L}$ is symmetric, and therefore an equivalence relation.*

Proof. Suppose there exists z with $x\mathcal{R}z$ and $z\mathcal{L}y$. From $x\mathcal{R}z$, we have $x = za$ and $z = xb$ for some $a, b \in S^1$. From $z\mathcal{L}y$, we have $z = cy$ and $y = dz$ for some $c, d \in S^1$. We need to find an element w such that $x\mathcal{L}w$ and $w\mathcal{R}y$. The Lean proof shows that in the non-trivial case, the element dza can be used for w . $\square$

Lemma 6 (Closure of $\mathcal{D}$ under $\mathcal{R}$ and $\mathcal{L}$). *The $\mathcal{D}$ relation is closed under composition with $\mathcal{R}$ and $\mathcal{L}$. If $x\mathcal{D}y$ and $y\mathcal{L}z$, then $x\mathcal{D}z$. Similarly, if $x\mathcal{D}y$ and $y\mathcal{R}z$, then $x\mathcal{D}z$. This property is used to prove the transitivity of $\mathcal{D}$.*

Proof. Suppose $x\mathcal{D}y$ and $y\mathcal{L}z$. By definition of $\mathcal{D}$, there exists an element w such that $x\mathcal{R}w$ and $w\mathcal{L}y$. Since $\mathcal{L}$ is an equivalence relation, from $w\mathcal{L}y$ and $y\mathcal{L}z$, we can deduce $w\mathcal{L}z$. Now we have $x\mathcal{R}w$ and $w\mathcal{L}z$, which by definition means $x\mathcal{D}z$. A similar argument holds for closure under $\mathcal{R}$. If $x\mathcal{D}y$ and $y\mathcal{R}z$, we use the commutation of $\mathcal{R}$ and $\mathcal{L}$ (5). $x\mathcal{D}y$ means there is a w with $x\mathcal{L}w$ and $w\mathcal{R}y$. Since $\mathcal{R}$ is an equivalence relation, $w\mathcal{R}y$ and $y\mathcal{R}z$ implies $w\mathcal{R}z$. So we have $x\mathcal{L}w$ and $w\mathcal{R}z$, which means $x\mathcal{D}z$. $\square$

Chapter 3

Basic Properties of Green's Relations

Lemma 7 (Characterization of Elements Below Idempotents). *Let e be an idempotent element in a semigroup S . An element $x \in S$ is $\mathcal{R}$ -below e if and only if $x = ex$. Similarly, x is $\mathcal{L}$ -below e if and only if $x = xe$. It follows that x is $\mathcal{H}$ -below e if and only if both conditions hold, i.e., $x = ex$ and $x = xe$.*

Proof. For the $\mathcal{R}$ -preorder, if $x \leq_{\mathcal{R}} e$, then $x = ez$ for some $z \in S^1$. Since e is idempotent, $e = e^2$, so $ex = e(ez) = e^2z = ez = x$. Conversely, if $x = ex$, then $x \leq_{\mathcal{R}} e$ by definition. The argument for the $\mathcal{L}$ -preorder is analogous. The statement for $\mathcal{H}$ follows directly from the definitions. $\square$

Lemma 8 (Preservation of Green's Relations by Morphisms). *Green's relations are preserved under semigroup morphisms. Let $f : S \rightarrow T$ be a semigroup morphism. If two elements $x, y \in S$ are related by any of Green's preorders or equivalence relations, then their images $f(x), f(y)$ are related by the same relation in T .*

Proof. If $x \leq_{\mathcal{R}} y$, then $x = yz$ for some $z \in S^1$. Applying the morphism f gives $f(x) = f(yz) = f(y)f(z)$, so $f(x) \leq_{\mathcal{R}} f(y)$. This extends to the equivalence relation: if $x \mathcal{R} y$, then $x \leq_{\mathcal{R}} y$ and $y \leq_{\mathcal{R}} x$, which implies $f(x) \leq_{\mathcal{R}} f(y)$ and $f(y) \leq_{\mathcal{R}} f(x)$, so $f(x) \mathcal{R} f(y)$.

A similar argument holds for $\mathcal{L}$. The preservation of $\mathcal{J}$ and $\mathcal{H}$ follows from their definitions. For $\mathcal{D}$, if $x \mathcal{D} y$, there exists z such that $x \mathcal{R} z$ and $z \mathcal{L} y$. Then $f(x) \mathcal{R} f(z)$ and $f(z) \mathcal{L} f(y)$, which implies $f(x) \mathcal{D} f(y)$. $\square$

Chapter 4

Green's Relations in Finite Semigroups

Important lemma: (2).

Lemma 9 (Equivalence of D and J Relations in Finite Semigroups). *In a finite semigroup, the $\mathcal{D}$ and $\mathcal{J}$ relations are equivalent. That is, for any two elements $x, y \in S$, $x\mathcal{D}y$ if and only if $x\mathcal{J}y$.*

Proof. The forward direction, that $x\mathcal{D}y$ implies $x\mathcal{J}y$, holds in any semigroup, not just finite ones. This is because if $x\mathcal{D}y$, there exists z such that $x\mathcal{R}z$ and $z\mathcal{L}y$. These relations imply $x \leq_{\mathcal{J}} z$ and $z \leq_{\mathcal{J}} y$, and by transitivity, $x \leq_{\mathcal{J}} y$. A symmetric argument shows $y \leq_{\mathcal{J}} x$, so $x\mathcal{J}y$.

The reverse direction relies on the semigroup being finite. Suppose $x\mathcal{J}y$. Then there exist s, t, u, v such that $x = syt$ and $y = uxv$. Substituting, we get $x = s \cdot u \cdot x \cdot v \cdot t$, showing that x satisfies a sandwich equation. By 2, there exist positive integers k and ℓ such that $(su)^k x = x$ and $x(vt)^\ell = x$.

We use $z = xv$ as the intermediate element to show $x\mathcal{D}y$. We need to prove $x\mathcal{R}xv$ and $xv\mathcal{L}y$.

For $x\mathcal{R}xv$: clearly $xv \leq_{\mathcal{R}} x$ since $xv = x \cdot v$. For the reverse, we have $x = x(vt)^\ell = xv \cdot t(vt)^{\ell-1}$, so $x \leq_{\mathcal{R}} xv$. Thus $x\mathcal{R}xv$.

$xv\mathcal{L}y$ follows similarly.

Therefore $x\mathcal{R}xv$ and $xv\mathcal{L}y$, which means $x\mathcal{D}y$. □

Lemma 10 (J-Equivalence Strengthening Preorders). *In a finite semigroup, if two elements are $\mathcal{J}$ -equivalent, then a one-sided preorder implies the corresponding one-sided equivalence. Specifically, if $x\mathcal{J}y$ and $x \leq_{\mathcal{R}} y$, then $x\mathcal{R}y$. Similarly, if $x\mathcal{J}y$ and $x \leq_{\mathcal{L}} y$, then $x\mathcal{L}y$.*

Proof. Suppose $x\mathcal{J}y$ and $x \leq_{\mathcal{R}} y$. The preorder $x \leq_{\mathcal{R}} y$ means there exists z such that $x = yz$. If $z = 1$ (in the monoid extension), then $x = y$ and we are done. Otherwise, z is a proper element and $x = yz$.

We need to show $y \leq_{\mathcal{R}} x$, i.e., that $y = xt$ for some t . From $x\mathcal{J}y$, there exist u, v such that $y = uxv$. Substituting $x = yz$, we get $y = u \cdot y \cdot zv$, so y satisfies a sandwich equation. By 2, there exists a positive integer n such that $y(zv)^n = y$.

Now we have:

$$y = y(zv)^n = yz \cdot v(zv)^{n-1} = x \cdot v(zv)^{n-1}$$

Thus $y \leq_{\mathcal{R}} x$, and combined with $x \leq_{\mathcal{R}} y$, we get $x\mathcal{R}y$.

The argument for $\mathcal{L}$ is symmetric. □

Lemma 11 (H-Equivalence from Sandwiching). *In a finite semigroup, if an element x can be written as $x = uxv$ for some $u, v \in S$, then x is $\mathcal{H}$ -equivalent to both ux and xv .*

Proof. Recall that $x\mathcal{H}y$ if and only if $x\mathcal{R}y$ and $x\mathcal{L}y$. We prove both $x\mathcal{H}ux$ and $x\mathcal{H}xv$.

Showing $x\mathcal{H}ux$: We need $x\mathcal{R}ux$ and $x\mathcal{L}ux$.

For $x\mathcal{R}ux$: We have $ux \leq_{\mathcal{R}} x$ trivially (since $ux = u \cdot x$). From $x = uxv$, we see $x \leq_{\mathcal{R}} ux$ (via v). Also, $x\mathcal{J}ux$ since $x = ux \cdot v$ (so $x \leq_{\mathcal{J}} ux$) and $ux = u \cdot x \cdot 1$ gives $ux \leq_{\mathcal{J}} x$. By 10, the $\mathcal{R}$ -preorder strengthens to $x\mathcal{R}ux$.

For $x\mathcal{L}ux$: We have $x\mathcal{J}ux$ as shown above. Since $ux = u \cdot x$, we have $ux \leq_{\mathcal{L}} x$. By 10, this strengthens to $ux\mathcal{L}x$.

Showing $x\mathcal{H}xv$: We need $x\mathcal{R}xv$ and $x\mathcal{L}xv$.

For $x\mathcal{R}xv$: We have $xv \leq_{\mathcal{R}} x$ trivially. Also $x\mathcal{J}xv$ since $x = u \cdot xv$ (so $x \leq_{\mathcal{J}} xv$) and $xv \leq_{\mathcal{J}} x$. By 10, this strengthens to $x\mathcal{R}xv$.

For $x\mathcal{L}xv$: We have $x = uxv = u \cdot xv$, so $x \leq_{\mathcal{L}} xv$. Also, $xv = x \cdot v$ gives $xv \leq_{\mathcal{L}} x$. With $x\mathcal{J}xv$, by 10, we get $x\mathcal{L}xv$.

Therefore $x\mathcal{H}ux$ and $x\mathcal{H}xv$. □

Chapter 5

Green's Lemma

Lemma 12 (Green's Lemma). *Let S be a semigroup and suppose $x\mathcal{R}y$ with witnesses $u, v \in S$ such that $x \cdot v = y$ and $y \cdot u = x$. Then:*

1. *The map $\rho_v : w \mapsto w \cdot v$ is a bijection from the $\mathcal{L}$ -class of x to the $\mathcal{L}$ -class of y .*
2. *The map $\rho_u : w \mapsto w \cdot u$ is the inverse of ρ_v .*
3. *These bijections preserve $\mathcal{H}$ -classes: for any a, b in the $\mathcal{L}$ -class of x , we have $a\mathcal{H}b$ if and only if $a \cdot v\mathcal{H}b \cdot v$.*

Dually, if $x\mathcal{L}y$ with witnesses u, v such that $v \cdot x = y$ and $u \cdot y = x$, then left translation $\lambda_v : w \mapsto v \cdot w$ is a bijection from the $\mathcal{R}$ -class of x to the $\mathcal{R}$ -class of y , with inverse λ_u , and these bijections preserve $\mathcal{H}$ -classes.

Proof. We prove the result for $\mathcal{R}$ -equivalent elements; the dual statement for $\mathcal{L}$ -equivalent elements follows by a symmetric argument.

Suppose $x\mathcal{R}y$ with $x \cdot v = y$ and $y \cdot u = x$. We show that $\rho_v : w \mapsto w \cdot v$ is a bijection from $[x]_{\mathcal{L}}$ to $[y]_{\mathcal{L}}$ by establishing that it maps into the correct set, is injective, is surjective, and has ρ_u as its inverse.

Translation Identity. The key observation is that for any $w\mathcal{L}x$, right translation by v followed by u returns w to itself: $w \cdot v \cdot u = w$. To see this, since $w\mathcal{L}x$, there exists z such that $w = z \cdot x$. Then:

$$w \cdot v \cdot u = z \cdot x \cdot v \cdot u = z \cdot y \cdot u = z \cdot x = w$$

This identity is central to all parts of the proof.

Maps To $(\rho_v \text{ maps } [x]_{\mathcal{L}} \text{ to } [y]_{\mathcal{L}})$. Let $w\mathcal{L}x$. By 4, right multiplication preserves $\mathcal{L}$ -equivalence, so $w \cdot v\mathcal{L}x \cdot v = y$. Thus ρ_v maps the $\mathcal{L}$ -class of x into the $\mathcal{L}$ -class of y .

Injectivity. Suppose $w, z \in [x]_{\mathcal{L}}$ and $w \cdot v = z \cdot v$. By the translation identity, $w = w \cdot v \cdot u = z \cdot v \cdot u = z$. Thus ρ_v is injective on $[x]_{\mathcal{L}}$.

Surjectivity. Let $z \in [y]_{\mathcal{L}}$. We claim that $z \cdot u \in [x]_{\mathcal{L}}$ and $\rho_v(z \cdot u) = z$. First, since $z\mathcal{L}y$ and $y \cdot u = x$, by 4 we have $z \cdot u\mathcal{L}y \cdot u = x$. Second, by the translation identity (applied with the roles of u and v swapped, using $z\mathcal{L}y$), we have $(z \cdot u) \cdot v = z$. Thus ρ_v is surjective.

Inverse (ρ_u is the inverse of ρ_v). The translation identity shows that for $w \in [x]_{\mathcal{L}}$, we have $\rho_u(\rho_v(w)) = w \cdot v \cdot u = w$. Symmetrically, for $z \in [y]_{\mathcal{L}}$, we have $\rho_v(\rho_u(z)) = z \cdot u \cdot v = z$. Thus ρ_u and ρ_v are inverses on the respective $\mathcal{L}$ -classes.

Preservation of $\mathcal{H}$ -classes. Let $a, b \in [x]_{\mathcal{L}}$. We show $a\mathcal{H}b \Leftrightarrow a \cdot v\mathcal{H}b \cdot v$. Recall that $\mathcal{H} = \mathcal{R} \cap \mathcal{L}$.

For the $\mathcal{L}$ -part: by 4, $a\mathcal{L}b$ implies $a \cdot v\mathcal{L}b \cdot v$, and conversely, $a \cdot v\mathcal{L}b \cdot v$ implies $a \cdot v \cdot u\mathcal{L}b \cdot v \cdot u$, i.e., $a\mathcal{L}b$.

For the $\mathcal{R}$ -part: since $a, b \in [x]_{\mathcal{L}}$, we have $a \cdot v \cdot u = a$ and $b \cdot v \cdot u = b$. This means $a \cdot v\mathcal{R}a$ (witnessed by u) and $b\mathcal{R}b \cdot v$ (witnessed by u). Thus $a\mathcal{R}b$ if and only if $a \cdot v\mathcal{R}b \cdot v$, by transitivity through these intermediate equivalences.

Dual Version. For $x\mathcal{L}y$ with $v \cdot x = y$ and $u \cdot y = x$, the proof is entirely symmetric: left translation $\lambda_v : w \mapsto v \cdot w$ is a bijection from $[x]_{\mathcal{R}}$ to $[y]_{\mathcal{R}}$, with inverse λ_u , and preserves $\mathcal{H}$ -classes. $\square$

Chapter 6

Subgroups of Semigroups

In this chapter, we define the notion of a subgroup within a semigroup. Unlike the standard definition in group theory where subgroups are subsets of groups, here we consider subgroups as subsets of semigroups that themselves form groups under the inherited operation.

Definition 13 (Subgroup of a Semigroup). Let S be a semigroup. A *subgroup* of S is a subset $H \subseteq S$ equipped with:

1. A *carrier set* $H \subseteq S$;
2. **Closure under multiplication:** For all $a, b \in H$, we have $a \cdot b \in H$;
3. An *identity element* $e \in H$ such that:
 - $e \in H$ (the identity is in the carrier);
 - For all $x \in H$, $e \cdot x = x$ (left identity);
 - For all $x \in H$, $x \cdot e = x$ (right identity);
4. An *inverse function* $\text{inv} : S \rightarrow S$ such that:
 - For all $x \in H$, $\text{inv}(x) \in H$ (closure under inverses);
 - For all $x \in H$, $\text{inv}(x) \cdot x = e$ (left inverse);
 - For all $x \in H$, $x \cdot \text{inv}(x) = e$ (right inverse).

With these properties, the carrier H forms a group under the multiplication inherited from S .

Note that the identity element e is uniquely determined by the carrier set: if e_1 and e_2 are both identities for the same carrier H , then $e_1 = e_1 \cdot e_2 = e_2$. Similarly, the inverse function is uniquely determined by the carrier and identity.

Definition 14 (Maximal Subgroup). A subgroup H of a semigroup S is called *maximal* if it is not properly contained in any other subgroup of S . That is, H is maximal if for every subgroup K of S with $H \subseteq K$, we have $H = K$.